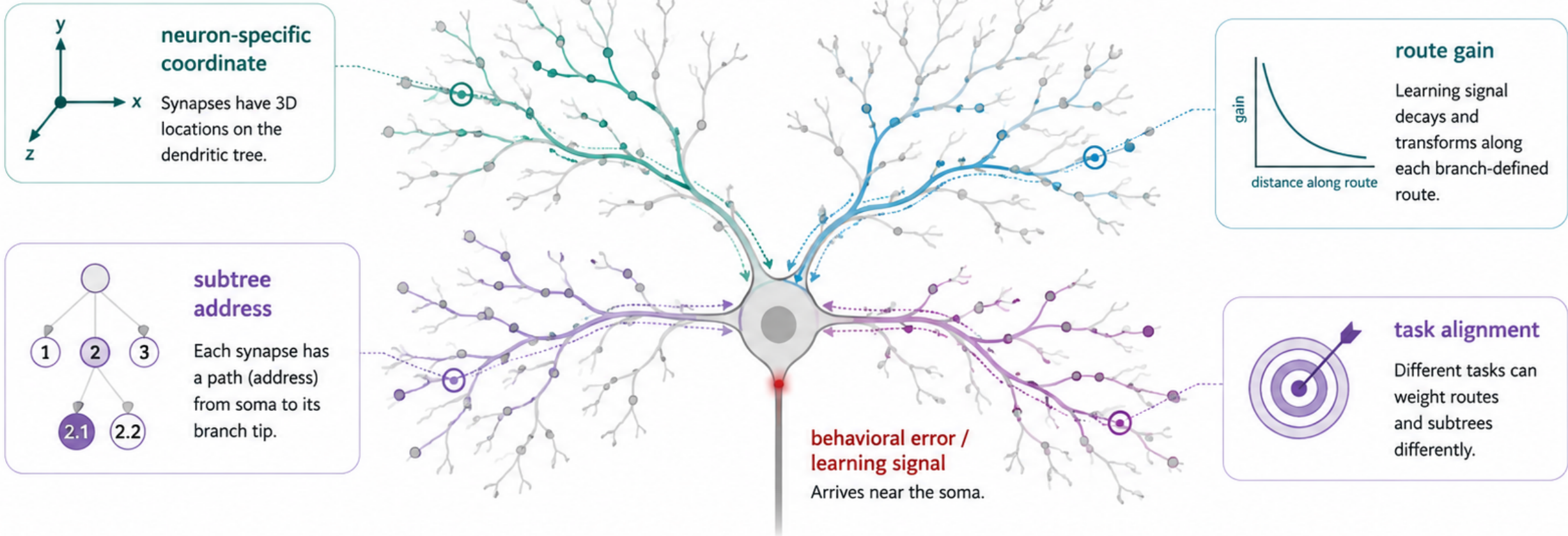


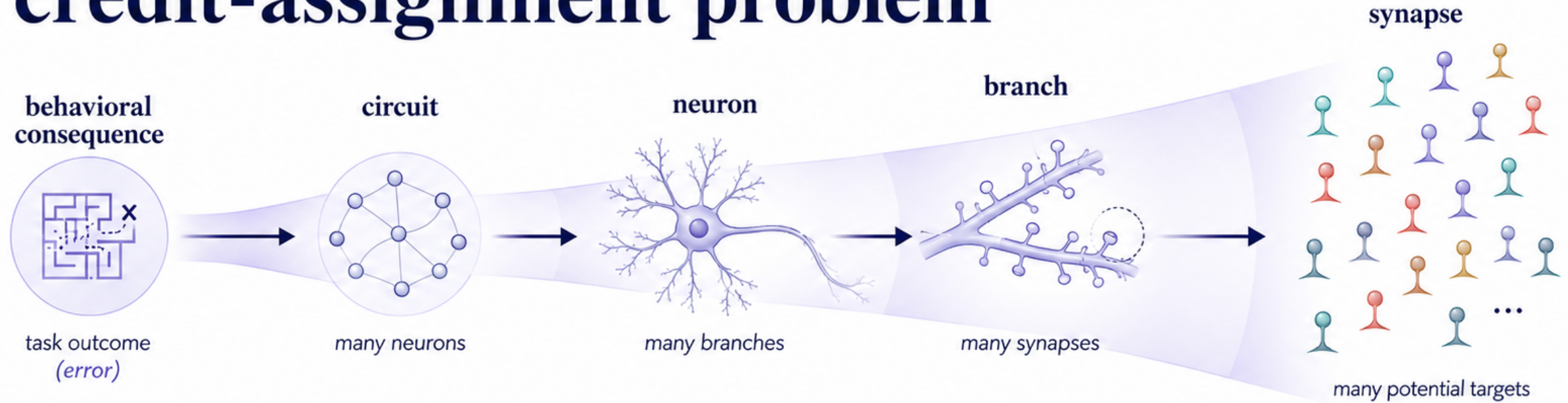
When dendritic structure helps local credit assignment

Coordinates, subtree addresses, route gain, and task alignment



Dendrites may help learning not just by computing, but by structuring *where credit goes* inside a neuron.

Learning in the brain is a credit-assignment problem

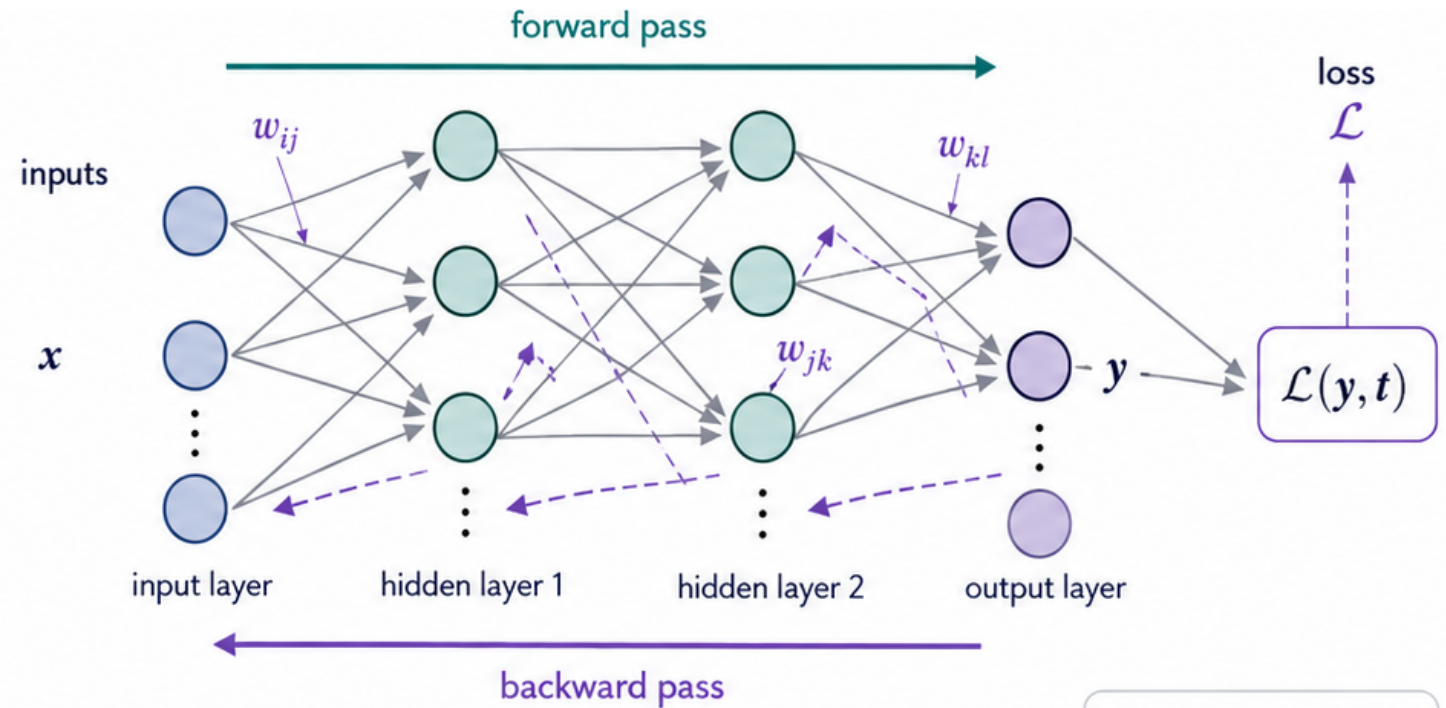


One behavioral error must be assigned to many local causes.



Sign, magnitude, and destination of credit are all needed.

Backpropagation solves parameter credit in artificial networks



$$\frac{\partial \mathcal{L}}{\partial w_{ij}} = \left(\frac{\partial \mathcal{L}}{\partial y_j} \right) \left(\frac{\partial y_j}{\partial w_{ij}} \right)$$



exact signed update



every parameter gets a destination

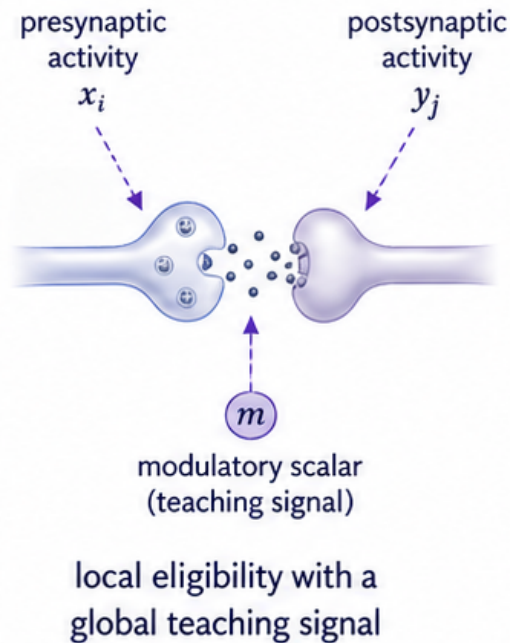


global coordination across layers

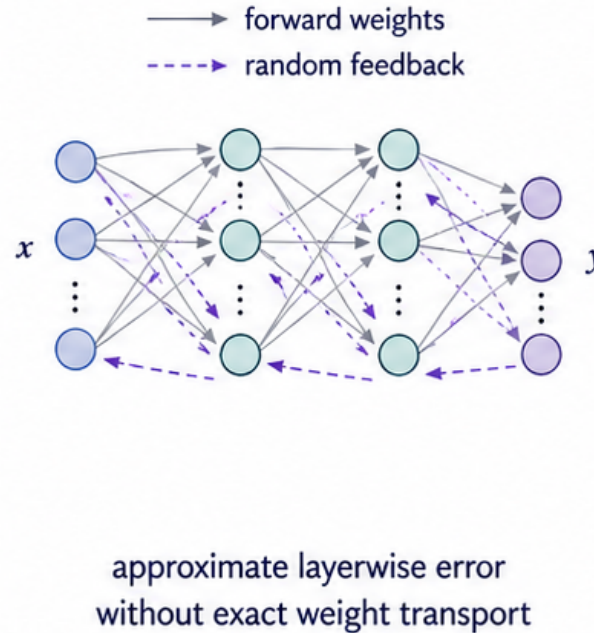
Backpropagation is the optimization reference because it tells every weight how it changed the loss.

Known biological learning theories solve different parts of credit assignment

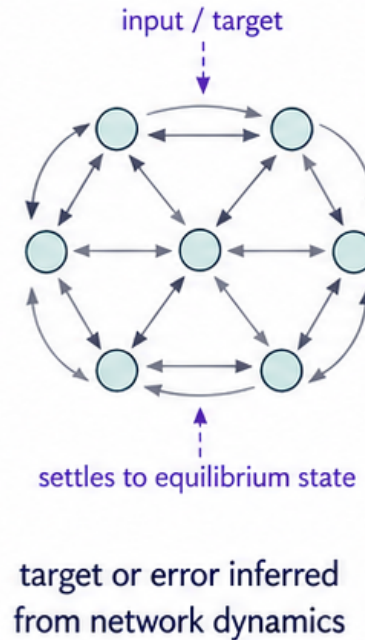
three-factor plasticity



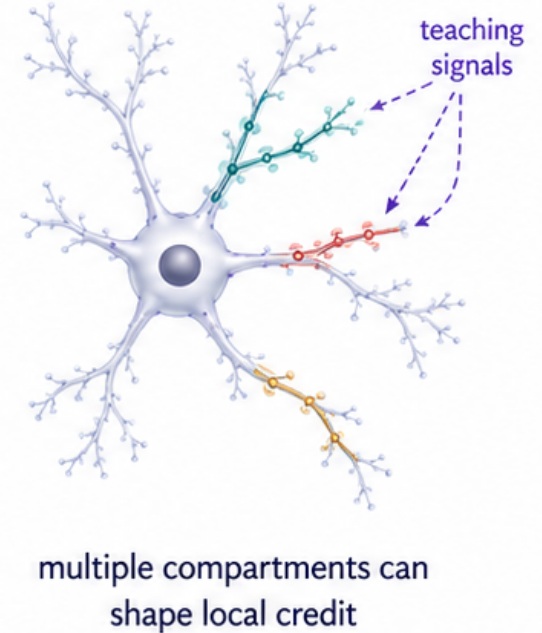
feedback alignment



predictive / equilibrium methods



dendritic / compartment models



Most approaches provide a neuron-level teaching coordinate or a few compartment signals; the fine within-tree destination problem remains open.

The unresolved problem begins inside the neuron

point neuron

many synapses
(each with weight w_i)



eligibility e_i

teaching signal δ_u

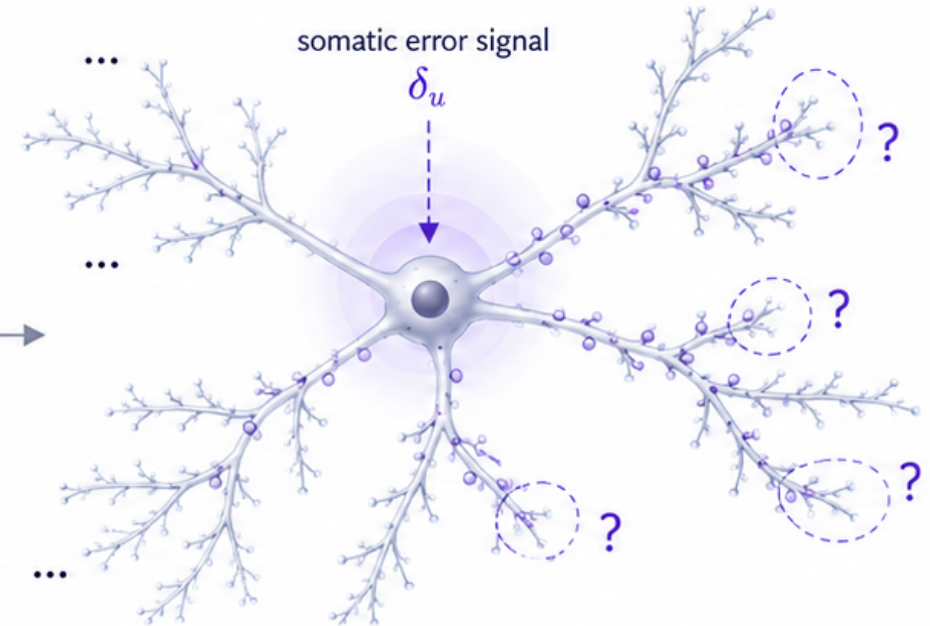
(one coordinate
for all synapses)

$$\frac{\partial \mathcal{L}}{\partial w_i} = e_i \delta_u$$

different local eligibility,
one shared task-dependent coordinate

same
behavioral error,
many possible
destinations

dendritic neuron



one neuron-level signal **does not specify**
which branch or synapse should change



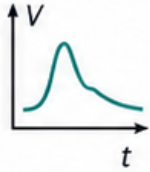
Within-neuron routing is the missing spatial step beyond neuron-level credit.

A dendritic tree adds local state, subtree addresses, and route gain

1 local state

Voltage/conductance dependence at compartments or branches.

$V, g \rightarrow$ nonlinear, state-dependent integration.



2 subtree address

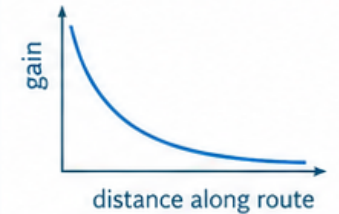
Soma-to-branch path hierarchy or ancestry route.

① → ② → ③ → ...

Encodes location and topology within the dendritic tree.

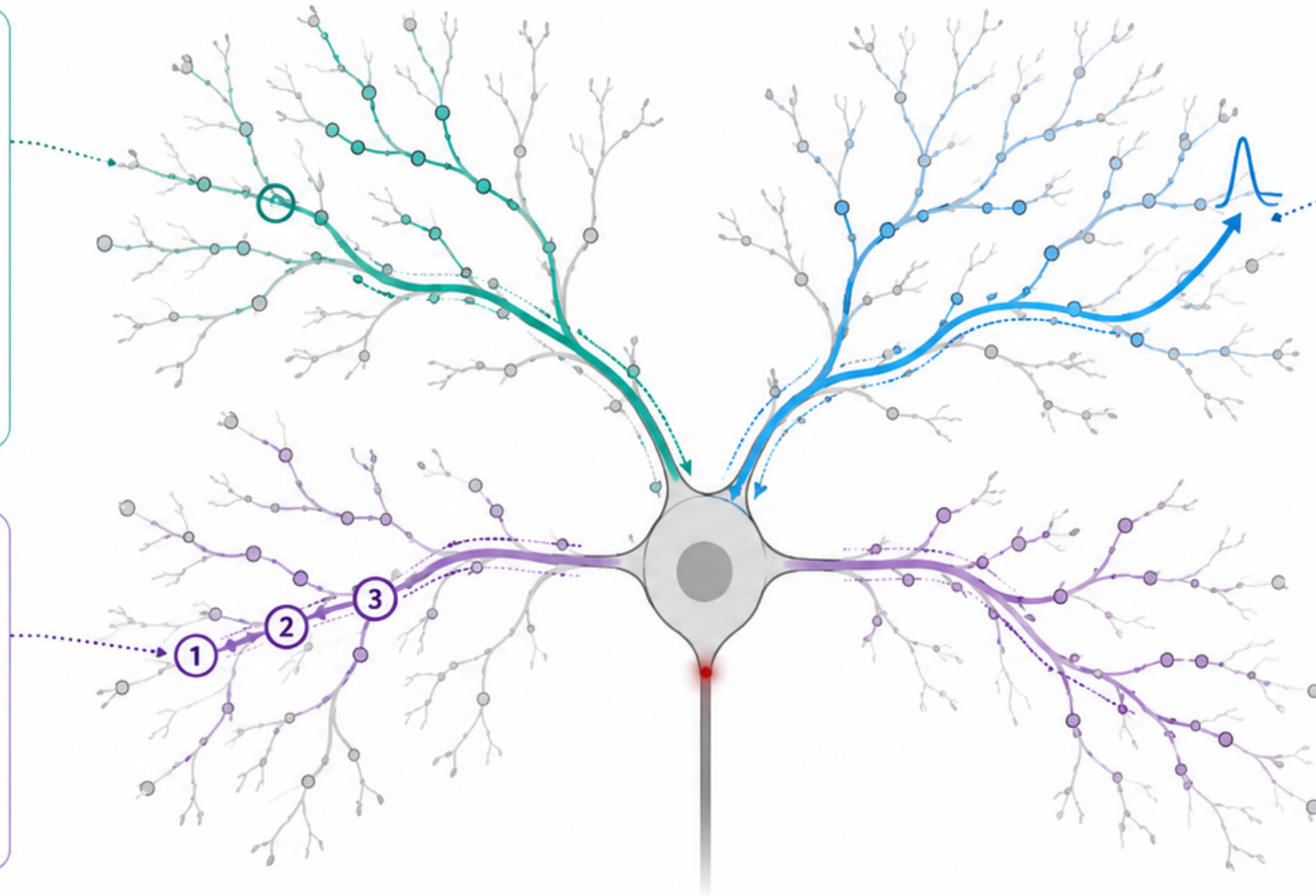
3 route gain

Attenuation or transformation of the signal along a branch-defined path.



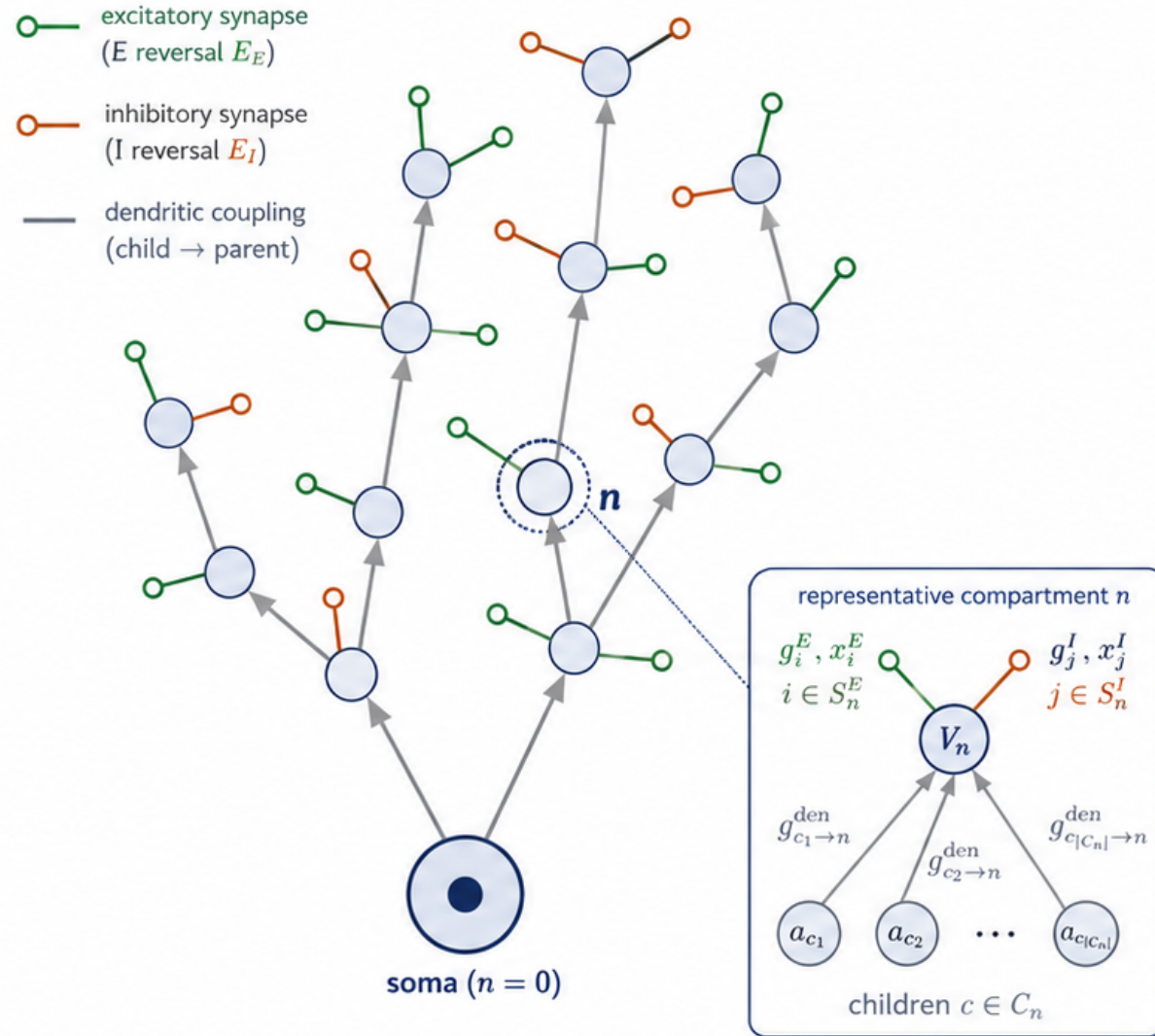
4 task alignment

Task demands determine whether local state, addresses, and route gain improve learning and performance.



Dendrites add spatial structure and state dependence beyond the point-neuron model.

A dendritic neuron is a conductance tree with explicit E/I synapses



1) Synapse sets and aggregates at compartment n

$$S_n^E, S_n^I, \quad S_n = S_n^E \uplus S_n^I$$

$$G_n^E = \sum_{i \in S_n^E} g_i^E x_i^E$$

$$G_n^I = \sum_{j \in S_n^I} g_j^I x_j^I$$

$$G_n^{\text{child}} = \sum_{c \in C_n} g_{c \rightarrow n}^{\text{den}}$$

$$J_n^{\text{child}} = \sum_{c \in C_n} g_{c \rightarrow n}^{\text{den}} a_c$$

2) Current balance and steady-state solution

$$C_n \frac{dV_n}{dt} = g_n^L (E_L - V_n) + G_n^E (E_E - V_n) + G_n^I (E_I - V_n) + \sum_{c \in C_n} g_{c \rightarrow n}^{\text{den}} (a_c - V_n)$$

$$g_n^{\text{tot}} = g_n^L + G_n^E + G_n^I + G_n^{\text{child}} \quad R_n^{\text{tot}} = \frac{1}{g_n^{\text{tot}}}$$

$$V_n = R_n^{\text{tot}} (g_n^L E_L + G_n^E E_E + G_n^I E_I + J_n^{\text{child}})$$

3) Normalized leaf approximation

$$g_n^L = 1, \quad E_L = E_I = 0, \quad E_E = 1$$

$$V_{\text{leaf}} = \frac{G^E}{1 + G^E + G^I}$$

4) Interpretation

Excitatory and inhibitory synapses both add positive conductance to the denominator; their different reversal potentials determine the sign of current.

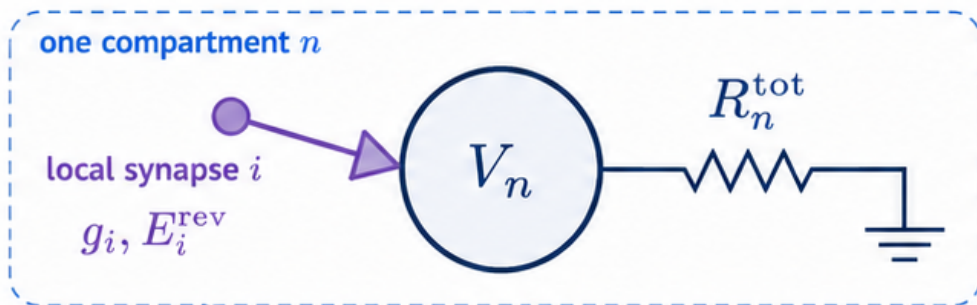
Exact local eligibility with explicit E/I synapses

Exact gradient factorization at synapse i on compartment n

$$\frac{\partial L}{\partial g_i} = e_i^{\text{den}} \delta_{n,u}^V$$

$$e_i^{\text{den}} = x_i R_n^{\text{tot}} (E_i^{\text{rev}} - V_n)$$

$$\delta_{n,u}^V = \frac{\partial L}{\partial V_n}$$



Local eligibility factors e_i^{den}

presynaptic activity

$$x_i^X$$

transmitter release
or spike-dependent
term

local input resistance

$$R_n^{\text{tot}}$$

total input resistance
seen at compartment n

driving force

$$(E_X - V_n)$$

difference between
reversal potential of
class X and local voltage

Sensitivity of V_n to a synaptic conductance of class X

$$\frac{\partial V_n}{\partial g_i^X} = x_i^X R_n^{\text{tot}} (E_X - V_n), \quad X \in \{E, I\}$$

Explicit excitatory and inhibitory gradient cases

$$\frac{\partial L}{\partial g_i^E} = x_i^E R_n^{\text{tot}} (E_E - V_n) \delta_{n,u}^V, \quad i \in \mathcal{S}_n^E$$

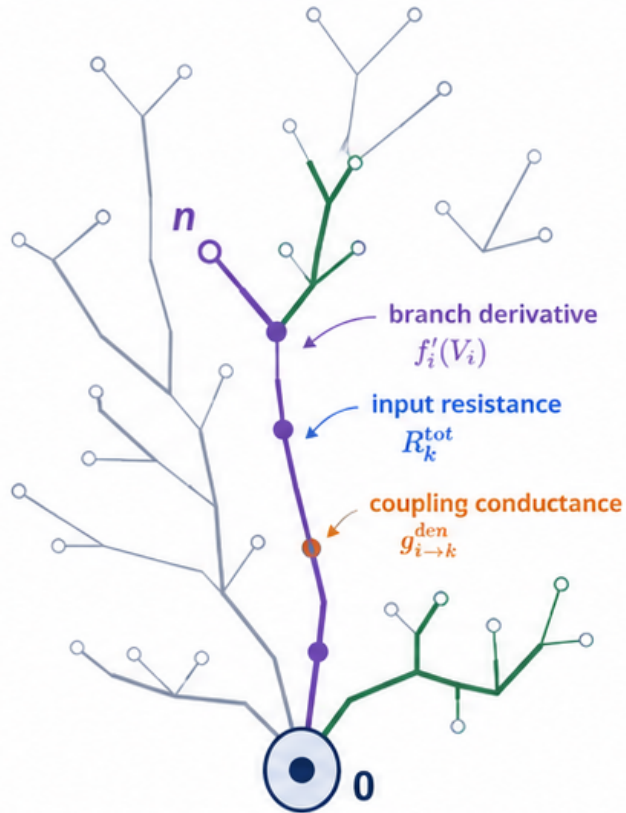
$$\frac{\partial L}{\partial g_j^I} = x_j^I R_n^{\text{tot}} (E_I - V_n) \delta_{n,u}^V, \quad j \in \mathcal{S}_n^I$$

Interpretation

- E and I have opposite local voltage sensitivities when $E_I < V_n < E_E$.
- The sign of the final loss gradient also depends on the compartment learning signal $\delta_{n,u}^V$.
- Near E_I , an inhibitory synapse can weakly drive its own current yet still change other gradients through conductance loading.

From tree transport to the general adjoint

Directed tree: single path transport of returned signal



Recursive transport along the unique path from n to 0

$$\delta_{0,u}^V = f'_0(V_0) \delta_u$$

$$\delta_{n,u}^V = \delta_{p(n),u}^V f'_n(V_n) R_{p(n)}^{\text{tot}} g_{n \rightarrow p(n)}^{\text{den}}$$

$$\delta_{n,u}^V = \delta_{0,u}^V \tilde{\alpha}_n$$

$$\tilde{\alpha}_n = \prod_{(i \rightarrow k) \in \text{path}(n \rightarrow 0)} f'_i(V_i) R_k^{\text{tot}} g_{i \rightarrow k}^{\text{den}}$$

$f'_i(V_i)$: branch derivative

R_k^{tot} : input resistance

$g_{i \rightarrow k}^{\text{den}}$: coupling conductance

$p(n)$ parent of node n on the path to the root
 $\text{path}(n \rightarrow 0)$ ordered set of directed edges from n to root 0
 $(n = i_0 \rightarrow i_1 \rightarrow \dots \rightarrow i_m = 0)$

General compartmental system: steady-state adjoint

$$F(V, g, x) = 0$$

steady-state residual
(nonlinear in general)

$$J_V = \frac{\partial F}{\partial V}$$

Jacobian w.r.t. V

$$J_V^T q = \nabla_V L$$

adjoint linear system

$$\frac{\partial L}{\partial g_i} = -q^T \frac{\partial F}{\partial g_i}$$

gradient of scalar objective L
(adjoint sensitivity formula)

For the standard residual convention

$$F = GV - b$$

nodal conductance matrix G , source b

$$\frac{\partial L}{\partial g_i} = x_i (E_i^{\text{rev}} - V_n) q_n$$

local sensitivity (synapse i on compartment n)

Relationship to directed-tree transport

$$q_n = R_n^{\text{tot}} \delta_{n,u}^V$$

adjoint equals input resistance
times returned signal

In a directed tree, error transport is a single path product; in a general compartmental system, it is carried by the full adjoint / Green's function.

Restricted routed local learning rule

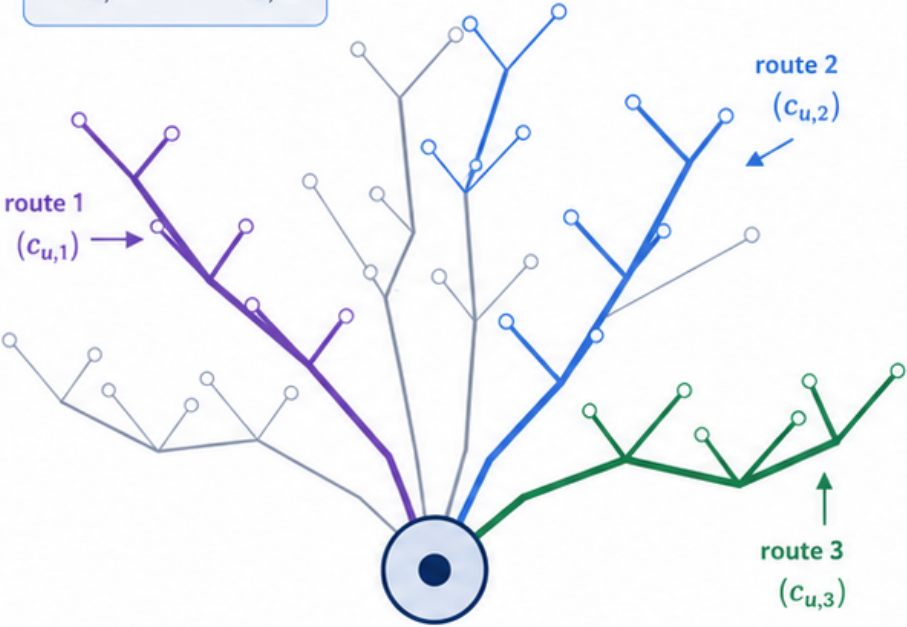
Exact and approximate local rules

Local approximate-gradient form:

$$\hat{\nabla}_{g_i} L = x_i R_n^{tot} (E_i^{rev} - V_n) \hat{\delta}_{n,u}^V$$

Exactness condition:

$$\hat{\delta}_{n,u}^V = \hat{\delta}_{n,u}^V$$



Routed teaching field (approximate local feedback field)

$$\hat{\delta}_{n,u}^V = \sum_{k=1}^K A_{u,nk} c_{u,k}$$

$A_u \in \mathbb{R}^{N_u \times K}$ fixed route/address matrix

$c_u = (c_{u,1}, \dots, c_{u,K})^\top$ communicated coefficients

K feedback bandwidth (number of route coefficients)

N_u routed nonsomatic compartments in neuron u

Address matrix A_u (compartments $\times$ routes)

	route 1	route 2	...	route K
compartment 1	1	0	...	0
compartment 2	1	0	...	0
compartment 3	0	1	...	0
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
compartment N_u	0	0	...	1

Route coefficients c_u (length K)

$$\begin{bmatrix} c_{u,1} \\ c_{u,2} \\ \vdots \\ c_{u,K} \end{bmatrix}$$

each value is assigned to the compartments of its route

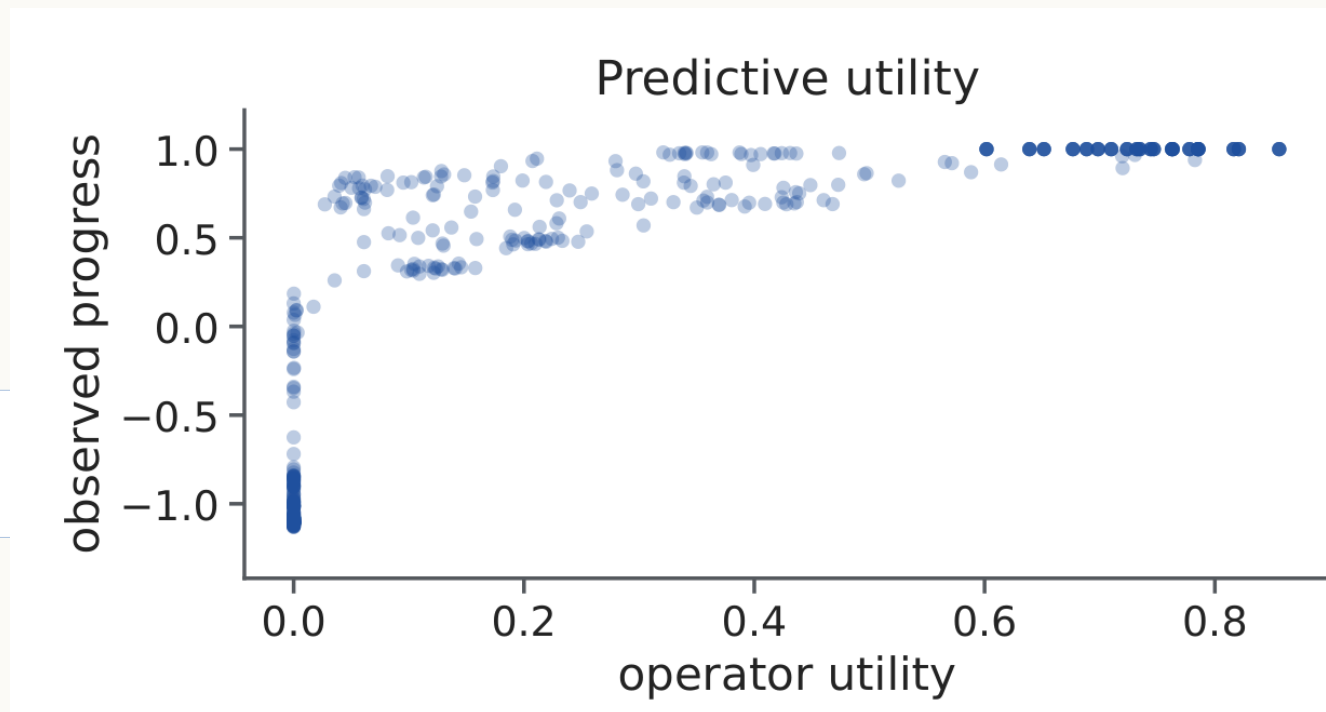
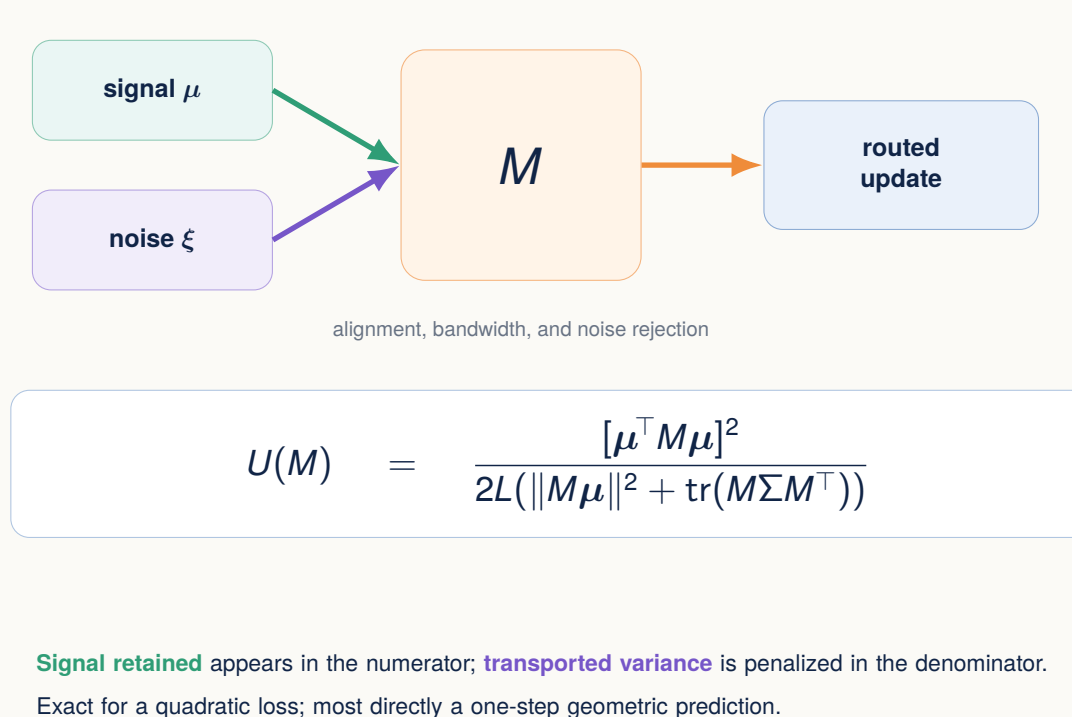
Bandwidth and routing choices: from scalar to exact compartment field

1) Layer-wide scalar	2) Neuron-specific feedback	3) Addressed routed field	4) Exact compartment field
one value shared by all neurons	one value per neuron, repeated within that neuron	several values per neuron, routed to selected subtrees	one independent value per compartment
bandwidth $K = 1$ shared across layer	bandwidth $K = 1$ per neuron	bandwidth $K > 1$ per neuron (routed)	bandwidth $K = N_u$ (exact)

Interpretation

The dendritic eligibility remains exact and local; *learning quality depends on how well the available routed field captures the exact credit field.*

A stochastic credit operator predicts when restricted routes help



Spearman $\rho_s = 0.937$ one-step $\rho_s = 0.916$ final

540 controlled conditions

The phase theory predicts both benefits and nulls from the geometry of the available credit operator.

Neuron identity dominates standard tasks

On standard tasks, the major bottleneck for local credit assignment is selecting the correct neuron, not resolving the exact path.

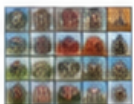
Tasks and setup

Two standard tasks

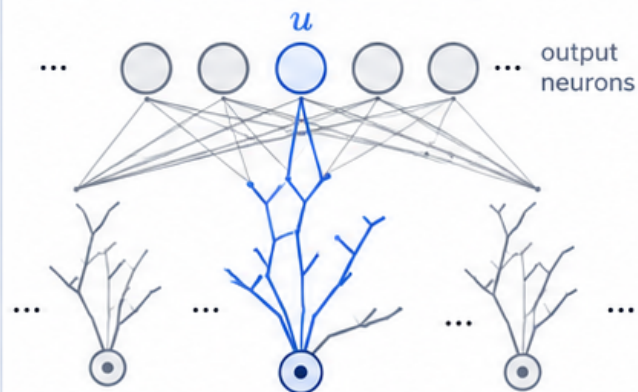
MNIST



Flattened CIFAR-10



Network schematic (final layer)



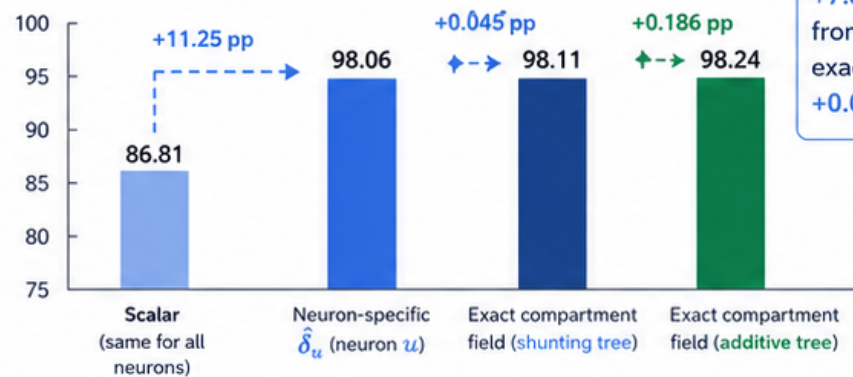
u = target neuron

Training signal must be routed back into neuron u .

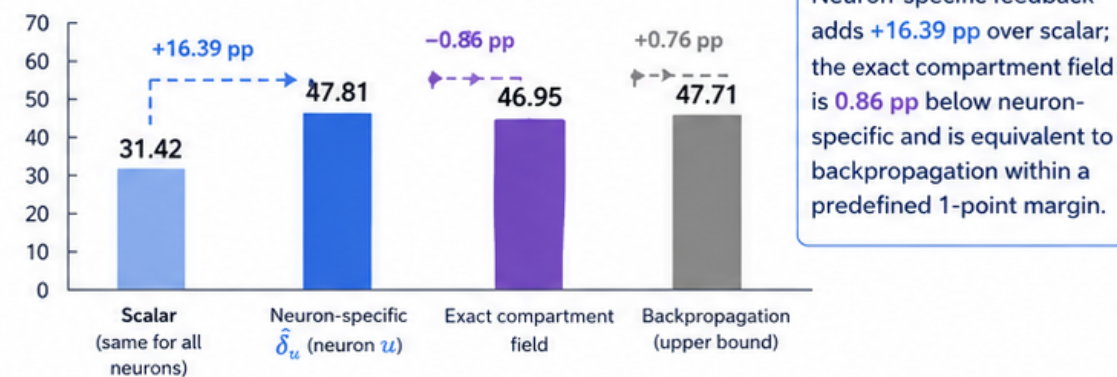
Compare scalar, neuron-specific, and exact compartment feedback under matched forward models.

Performance comparison (higher is better)

MNIST (test accuracy, %)



Flattened CIFAR-10 (test accuracy, %)



All results use matched forward models and identical training settings.

Takeaways

- Correct **neuron identity** is the main missing coordinate.
- Exact path resolution **matters little** on standard tasks.
- This motivates tasks with **within-neuron credit conflict**.

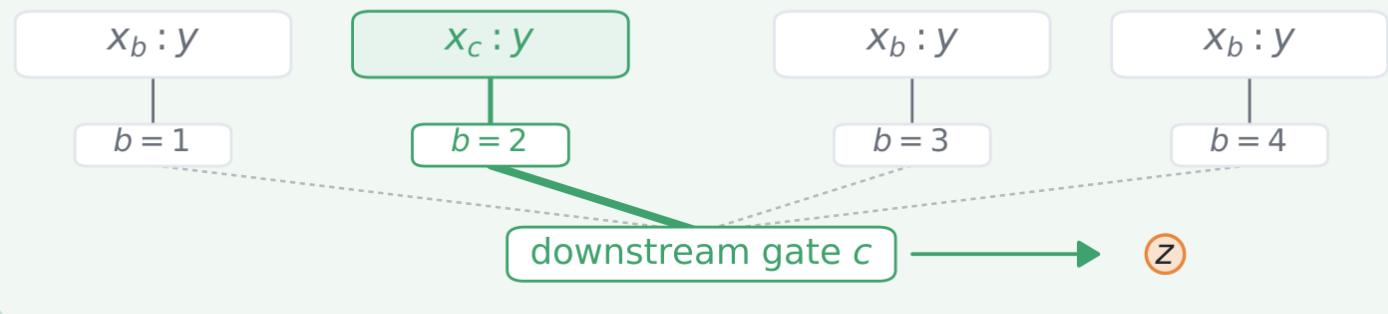
Notation

u = target neuron (the correct output neuron for the training example).

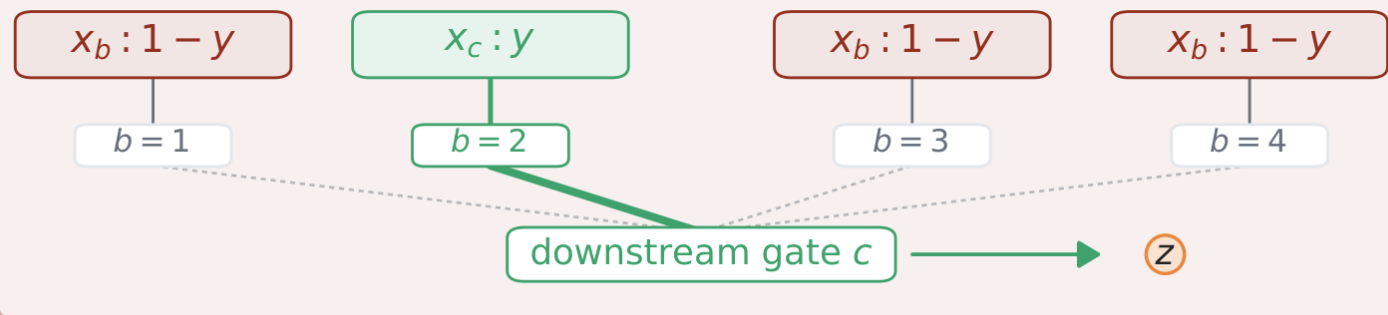
$\hat{\delta}_u$ = neuron-specific learning signal for neuron u , identical across all compartments of u .

Credit conflict creates a demand for branch-specific signals

compatible, $\chi = 0$



conflicting, $\chi = 1$



Constructed stress test. All B branches receive input. Context c selects the branch that determines the label. Each nonselected branch carries opposite-class evidence with probability χ .

$$\lambda_{\text{shared}} = B - 2\chi(B - 1)$$
$$\chi_c = \frac{B}{2(B - 1)}$$

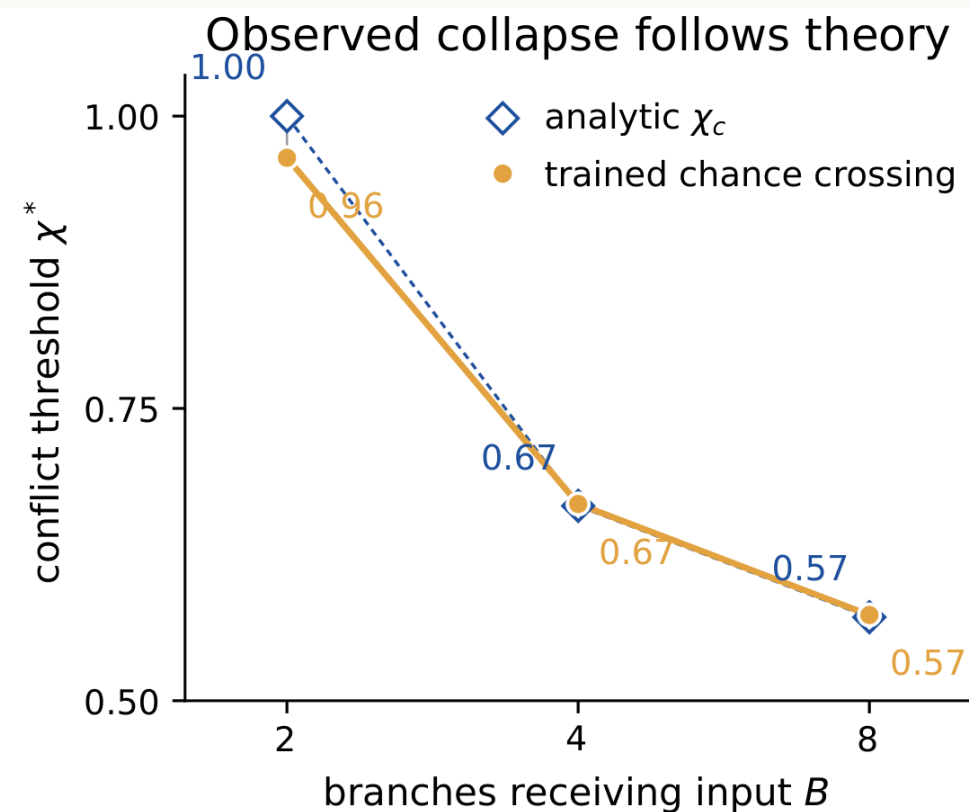
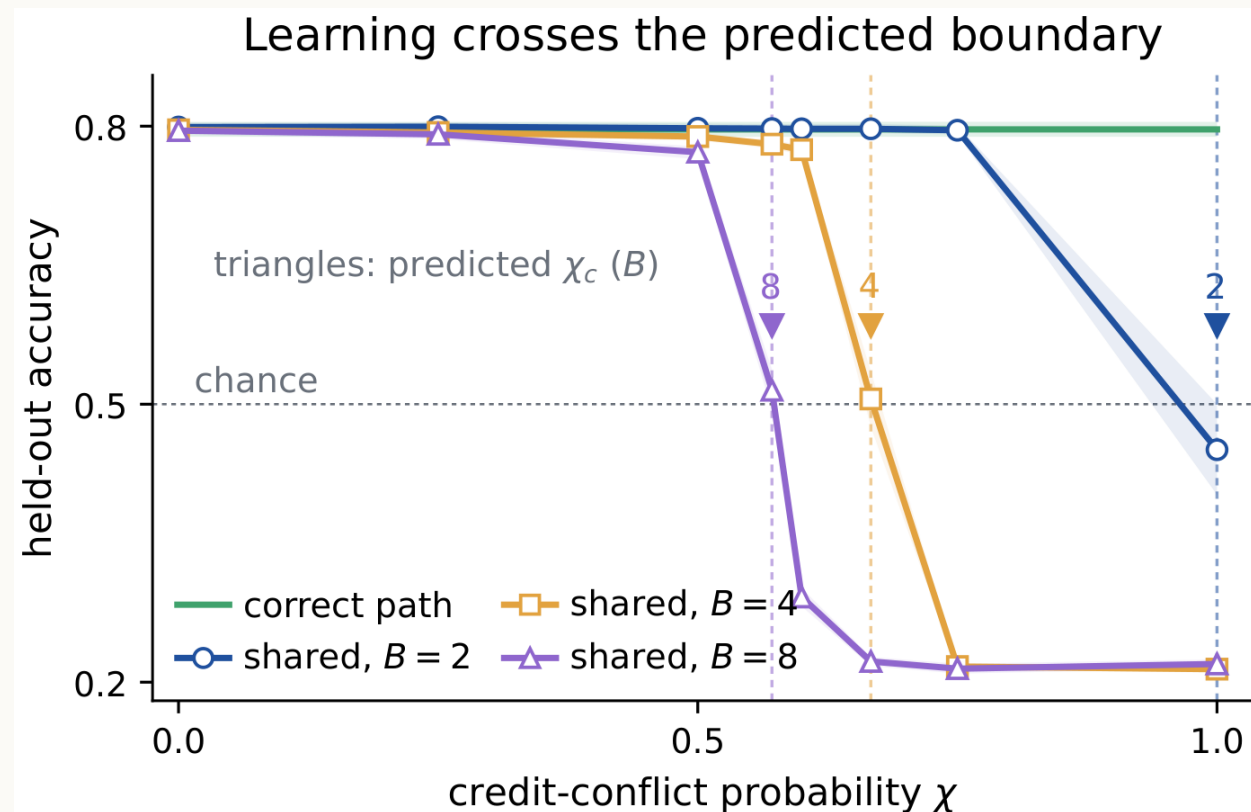
Correct route: sends the returned signal only to branch c .

Shared signal: sums incompatible branch updates and becomes noninformative above χ_c .

Mechanism-matched existence test, not a claim about every natural task.

The task isolates a need for an internal branch address when one neuron must assign opposing credit to sibling branches.

Shared credit fails at the predicted conflict boundary



$$\chi_c = \frac{B}{2(B-1)}$$

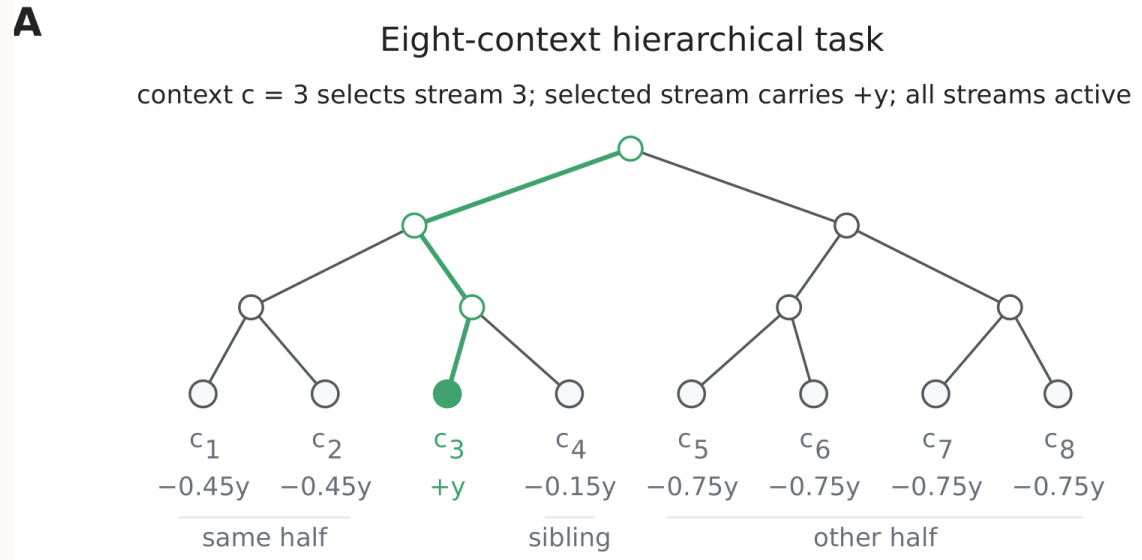
analytic and trained crossings agree for $B \in \{2, 4, 8\}$

35–58 pp
correct branch route over a neuron-shared signal at full conflict

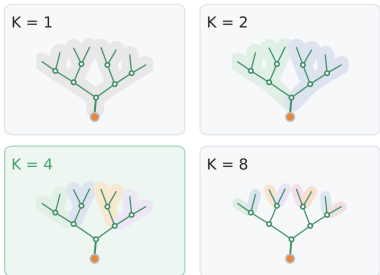
Ownership control. Cyclic derangement preserves rank and sparsity but sends each signal to the wrong branch; it fails across 20 seeds per B .

Branch-specific credit becomes useful precisely when a shared neuronal signal cancels across competing branches.

The right subtree address matters; fine topology adds a narrow gain

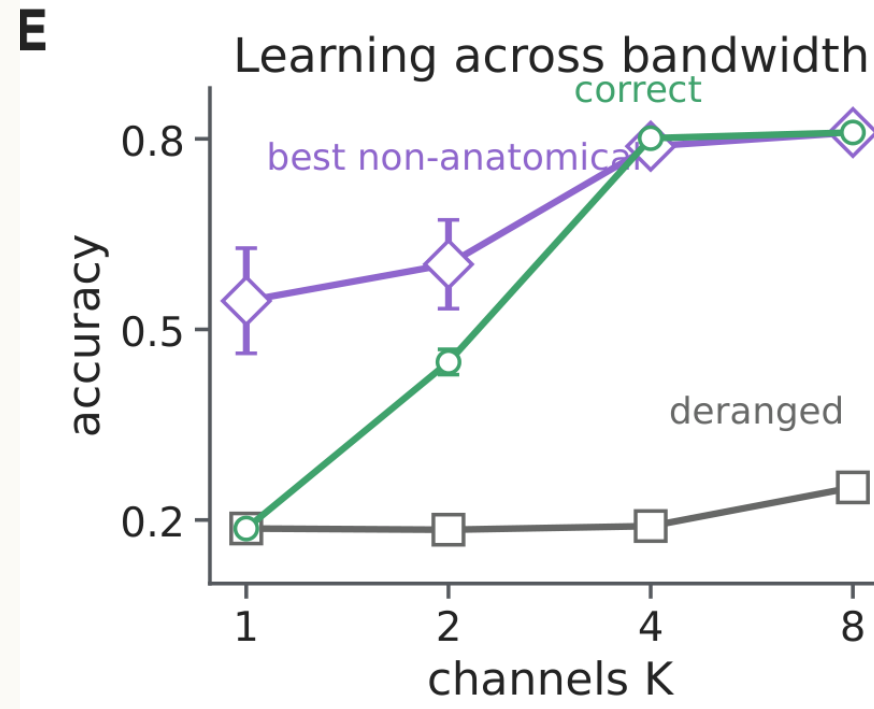


B Feedback bandwidth within one neuron



$K = 1, 2, 4, 8$ is the number of independent within-neuron route coefficients.

Route families are matched for rank, sparsity, forward computation, and feedback cost.



+61.1 pp

correct address vs. derangement at $K = 4$

+1.27 pp

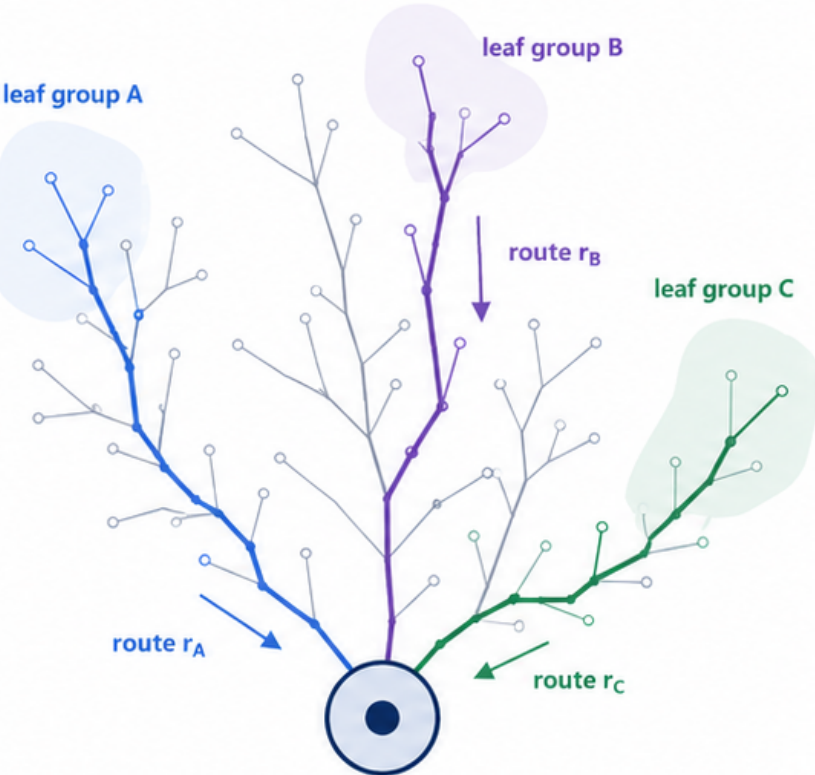
ancestry vs. strongest matched low-rank control at $K = 4$

Correct assignment carries most of the benefit. The nested anatomical basis adds only a small intermediate-bandwidth increment.

Addresses can be essential, while the extra benefit of this particular tree topology remains conditional and modest.

Real arbors provide sparse candidate routes

1) Reconstructed arbor with ancestry-defined routes



Measured anatomy, modeled credit fields.

2) Route capture: definition and results (8 channels)

Capture of a route dictionary A

$$C_A(q) = \frac{\|P_A q\|^2}{\|q\|^2}$$

Capture is the fraction of modeled credit-field energy contained in the span of the route dictionary A .

Results at 8 channels

Subtree routes retain of dense rank-matched capture	85%
Using about of dense route-matrix connections	6.9%
Dense capture per connection (subtree vs. dense)	14.2×
Advantage over shuffled dictionary at matched density	≈ 2.7×

Dense: rank-matched optimal dense dictionary.
Shuffled: random dictionary with the same density.

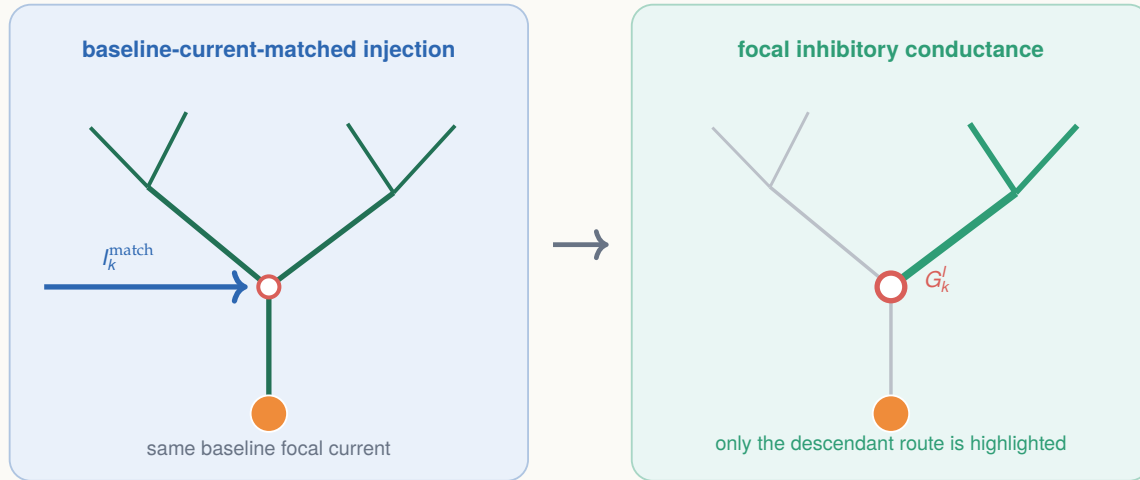
3) Interpretation

- Reconstructed arbors provide sparse candidate route dictionaries.
Ancestry-defined subtree routes capture most of the modeled credit-field energy while using very few potential routing connections.
- These are modeled fields on measured anatomy, not observed task gradients.
Capture is computed from operator-induced credit fields on the reconstructed morphology. No task gradients were used.
- Ordering replicates in a disjoint 47-cell cohort.
The same ordering (subtree > dense at matched density > shuffled) was observed in a held-out set of 47 cells with no cells shared across partitions.

Same ordering in a descriptive 10-cell second-mouse cohort

Across a disjoint 47-cell cohort, real arbors yield dictionaries that are far sparser yet much more capture-efficient than random alternatives.

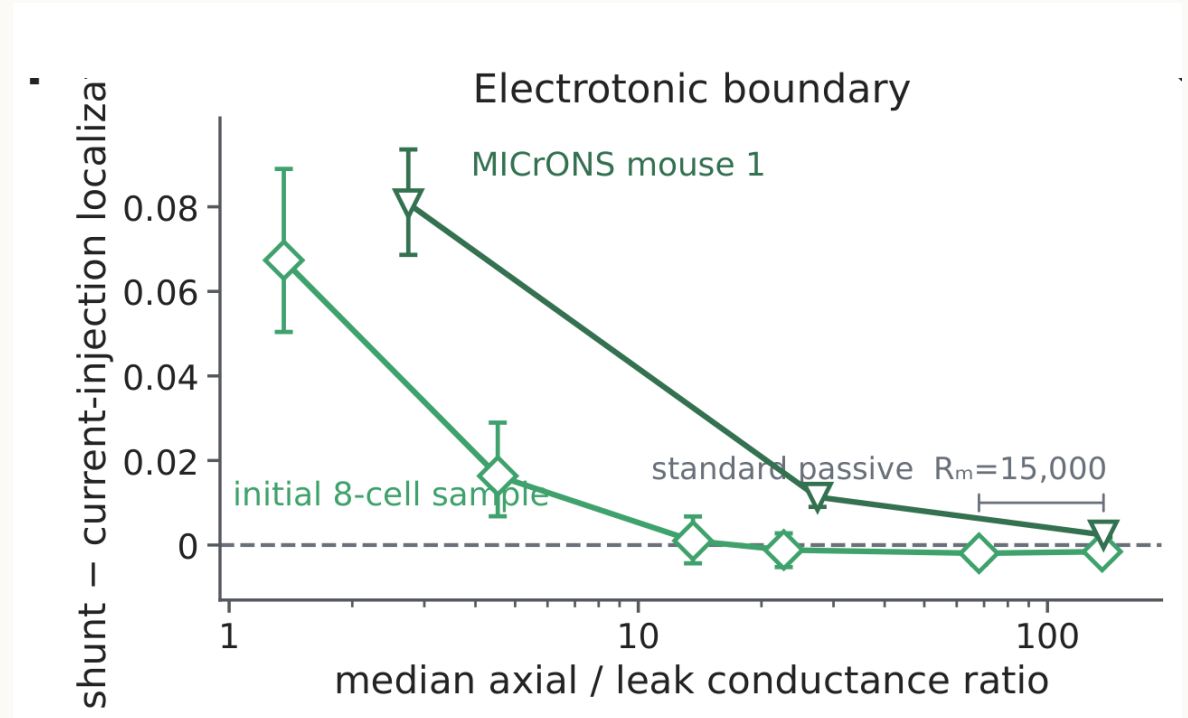
Focal shunting changes descendant credit only in permissive regimes



$$\frac{\partial \log \tilde{\alpha}_n^{\text{cond}}}{\partial G_k^I} = -R_k^{\text{tot}} \mathbf{1}[k \in \mathcal{A}(n)]$$

first-order change is restricted to descendants whose route passes through site k

Somatic voltage is restored after both interventions. Only the shunt changes the conductance matrix G and therefore the adjoint field $q = G^{-\top} \nabla_V \mathcal{L}$.



Standard passive calibration ($R_m = 15,000$): null.

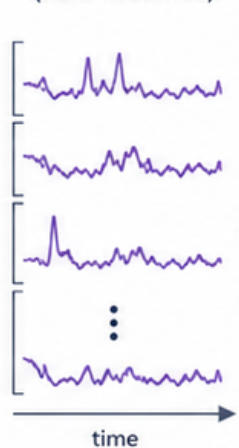
Lower resistance / background conductance: descendant localization emerges.
Conditional route-gain mechanism; not a general trained-learning advantage.

Shunting can regulate an already-addressed credit route, but ordinary passive parameters do not guarantee the effect.

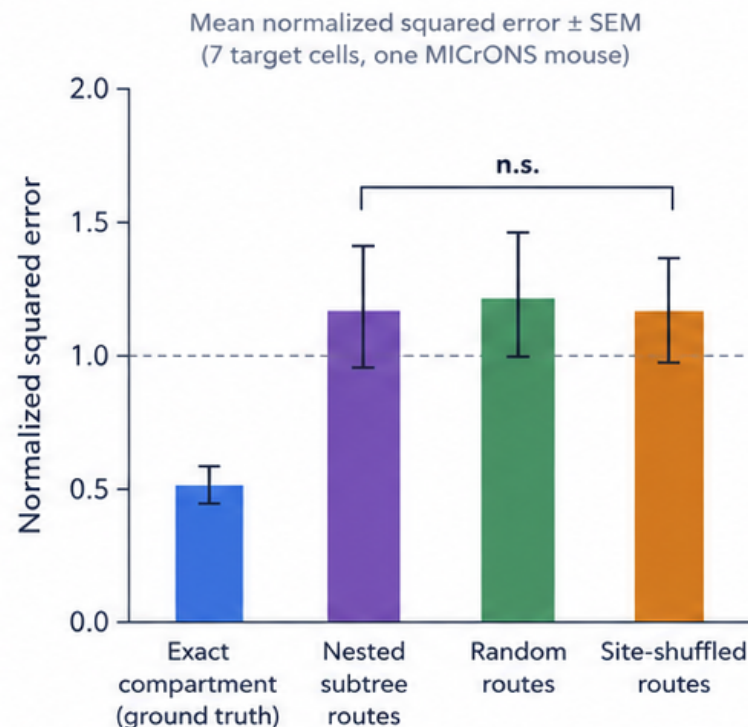
Measured visual responses show no morphology-specific alignment

MICrONS response-prediction setup

Presynaptic visual responses
(held-in stimuli)



Held-out prediction error (lower is better)



Across 7 target cells from one MICrONS mouse, nested subtree routes do not beat random or site-shuffled routes for held-out prediction.

n.s. $p > 0.05$ (paired t-test vs. random routes)

Interpretation

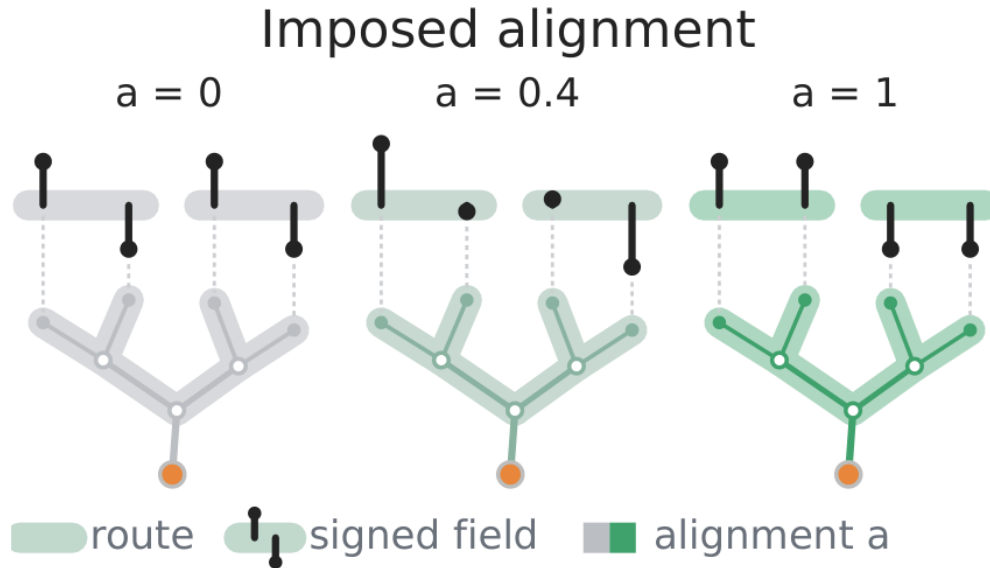
No significant morphology-specific advantage in held-out prediction.

Also no advantage in response-derived field capture or within-arbor structure-function similarity.

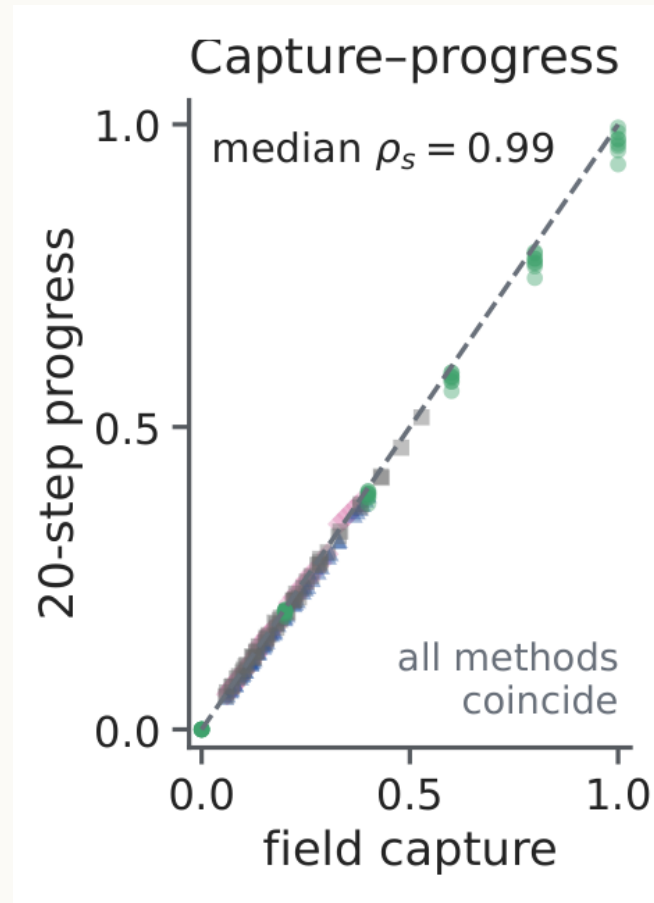
This is the strongest boundary on the biological claim.

Imposed alignment makes the same anatomical dictionary useful

$$\phi(a) = \sqrt{a} u_{\parallel} + \sqrt{1-a} u_{\perp}, \quad C_A[\phi(a)] = a$$



Anatomy, routes, field energy, and curvature stay fixed. Only the imposed task-credit field rotates into the route span.



8/8

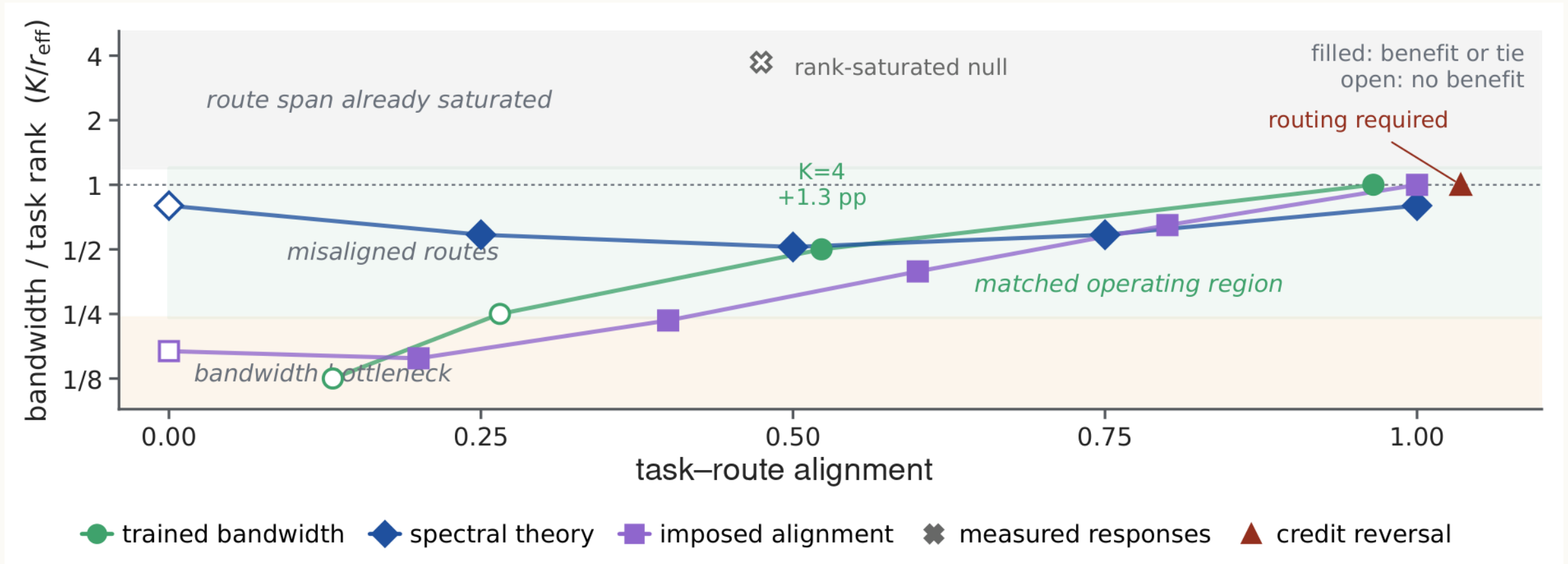
lose at $a = 0$;
win at $a = 1$

$\rho_s = 0.99$
capture-progress

Establishes: representational sufficiency of a fixed route dictionary under controlled alignment.
Does not establish: trained or endogenous biological alignment.

Morphology becomes useful when the task-credit field lies in the span of its available routes.

Dendritic benefit lies in a matched alignment–bandwidth regime



theoretical regime map; boundaries are not fitted

1 Coordinate

Neuron identity closes most of the standard-task gap.

2 Address

Routes help when local credits conflict and the task matches their span.

3 Gain

Anatomy is sparse; shunting regulates routes only in permissive regimes.

Dendritic structure is a conditional substrate for routing local credit—not a general replacement for backpropagation.